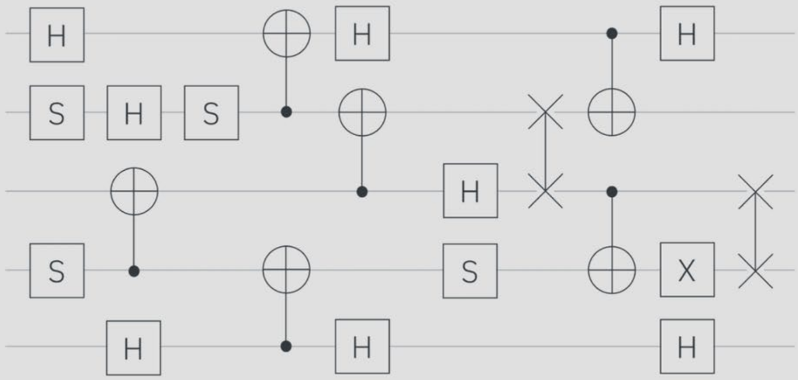
IBM **Quantum**

Road to Practitioner



Lecture topics and links

Orientation and Unit 1 (lecture link) - May 6, 2026
Topics Covered: Intro to quantum computing, towards quantum advantage, Intro to Qiskit SDK, Hello World, Qiskit Patterns, Scaling the GHZ state to 100 qubits, Encodings and Ansatzes, IQP platform and manage jobs

Unit 2 Lecture 1 (lecture link) – May 11, 2026
Topics Covered: Overview of Variational algorithms, Case studies: VQE, QAOA

Unit 2 Lecture 2 (lecture link) – May 13, 2026
Topics Covered: Overview of QML and data encoding methods, Quantum Kernel Estimation

Unit 3 Lecture 1 (lecture link) – May 20, 2026
Topics Covered: Optimizing circuits: Transpilation, Preset and Custom Pass Manager

Unit 3 Lecture 2 (lecture link) – May 27, 2026
Topics Covered: Dealing with noise: Overview of different ES and EM techniques (DD, Twirling, REM, ZNE, PEC), comparing results different ES and EM techniques (DD, Twirling, REM, ZNE)

Unit 3 Lecture 3 (lecture link) – June 3, 2026
Topics Covered: Dynamic Circuits and other Advanced Techniques, Hands-on lab

Unit 4 Lecture 1 (lecture link) – June 10, 2026
Topics Covered: Simulating the dynamic of a Ising model (spin chain + Trotterization), Sample based methods: SQD/KQD/SKQD

Unit 4 Lecture 2 (Final lecture link) – June 17, 2026
Topics Covered: Quantum Centric Supercomputing, Resource management (Slurm), introducing the Capstone Project

Capstone Project info and deadlines to follow